Question	Answer	Marks	Partial Marks
6(b)	$\frac{2-x}{x}$ or $\frac{2}{x}-1$ final answer	**3**	**M1** for $y(x + 1) = 2$ or $x = \frac{2}{y+1}$ or better **M1** for $\frac{2-y}{y}$ or $xy = 2 - x$ oe
6c	$-\frac{5}{6}$ –0.833 or better	**2**	**M1** for $[64^x =]\ 2^{6x}$ or $(2^6)^x$ or $6x = -5$
6(d)	$\frac{7-9x}{(5x-3)(x+1)}$ or $\frac{7-9x}{5x^2+2x-3}$ or $-\frac{9x-7}{5x^2+2x-3}$ final answer	**4**	**B1** for $\frac{1}{5x-3}-\frac{2}{x+1}$ **M1** for $x + 1 - 2(5x - 3)$ seen isw **M1** for $(5x - 3)(x + 1)$ seen isw
7(a)	0, –1.5 oe, –2.8	**3**	**B1** for each
7(b)	Correct graph	**4**	**B3 FT** for 10 or 11 correct points FT *their* table or **B2 FT** for 8 or 9 correct points FT *their* table or **B1 FT** for 6 or 7 correct points FT *their* table
7(c)	65 to 67	**1**	**FT** intersection of *their* graph with $y = -2$
7(d)	$y = 2-\frac{x}{40}$ oe ruled	**M2**	**M1** for $[y =]\ 2-\frac{x}{40}$ oe soi or for $3\cos 2x = 2-\frac{x}{40}$ oe soi
	32 to 36	**B1**	
8(a)(i)	$\sin[BOC] = \frac{2}{6}$ or better oe	**M1**	
	19.47…	**A1**	
8(a)(ii)	64.6 or 64.55 to 64.58	**3**	**M2** for $\frac{360-135-19.5}{360}\times\pi\times 6^2$ oe or **M1** for $\frac{k}{360}\times\pi\times 6^2$ oe

Question	Answer	Marks	Partial Marks
8(a)(iii)	16.1 or 16 10 to 16.13	**5**	**M2** for $2 \times \sqrt{6^2 - 2^2}$ oe or $2 \times 6 \cos 19.5$ oe or **M1** for $OC^2 + 2^2 = 6^2$ oe or $6 \cos 19.5$ or better AND **M2** for $\sqrt{6^2 + \textit{their } OD^2 - 2 \times 6 \times \textit{their } OD \times \cos 135}$ OR **M1** for $6^2 + \textit{their } OD^2 - 2 \times 6 \times \textit{their } OD \times \cos 135$ **A1** for 259 to 260
8(a)(iv)	94.2 or 94.3 or 94.15 to 94.27… nfww	**4**	**M1** for $\frac{1}{2} \times 6 \times \textit{their } OD \times \sin 135$ oe **M1** for $\frac{1}{2} \times 6 \times 2 \times \sin(90 - 19.5)$ oe or for $\frac{1}{2} \times \textit{their } OC \times 2$ **M1dep** for *their* **(a)(ii)** + *their* two triangle areas
8(b)	1000 cao	**3**	**M2** for $160 \times \left(\frac{20}{8}\right)^2$ or $160 \div \left(\frac{8}{20}\right)^2$ oe or **M1** for $\left(\frac{20}{8}\right)^2$ or $\left(\frac{8}{20}\right)^2$ oe OR **M2** for $\frac{\text{sector angle}}{360} \times \pi 20^2$ or **M1** for $\frac{160}{\pi 8^2} \times 360$ oe or better OR **M2** for $\frac{\text{percentage}}{100} \times \pi 20^2$ oe or better or **M1** for $\frac{160}{\pi 8^2}$ [× 100] oe or better
9(a)	[AB^2 =] $(3 - 0)^2 + (3 - 2)^2$ oe or better	**M1**	or $\begin{pmatrix} 3 \\ 1 \end{pmatrix}$ oe
	[AC^2 =] $(0 - 2)^2 + (4 - 0)^2$ oe or better	**M1**	or $\begin{pmatrix} 4 \\ -2 \end{pmatrix}$ oe
	[BC^2 =] $(0 - 3)^2 + (4 - 3)^2$ oe or better	**M1**	or $\begin{pmatrix} 1 \\ -3 \end{pmatrix}$ oe